Graph Neural Networks for Video Anomaly Detection

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Abstract

This technical report describes a comprehensive analysis on how Dynamic Graph Neural Networks can be used for video anomaly detection in traffic surveillance situations. To address this problem, we design a Temporal Graph Neural Network (Temporal DGNN) that employs spatial GCNs on a sequence of per-frame graphs followed by temporal GRU modeling. In our experiments, we use the NiAD_large_graphs dataset, consisting of 1,176 video samples (842 for training, 128 for validation, 206 for test) originating from the NiAD (Night-time Accident Detection) dataset. The pipeline translates 30 frames of video data into graph objects using PyTorch Geometric, with a different number of nodes per frame determined by vehicle detection. Data augmentation (brightness, flip, spatial shift) is performed on the anomalous class in order to address issues of class imbalance. The Temporal DGNN reaches a test accuracy of **89.32%** with an **AUC-ROC of 0.81**, showing effective learning of spatial-temporal patterns for anomaly detection. This report discusses the complete data pipeline from video processing to graph formation, mathematical foundations of GNNs, Temporal DGNN architecture, and experimental results.

Keywords: Graph Neural Networks, Video Anomaly Detection, Dynamic Graphs, Traffic Surveillance, Deep Learning, Temporal Modeling, PyTorch Geometric

Contents

1 Introduction **5**
1.1 Research Problem . . . 5
1.2 Dataset: NiAD_large_graphs . . . 5
1.3 Research Contributions . . . 6
1.4 Report Organization . . . 6
1.5 System Architecture Overview . . . 6

2 Mathematical Background **7**
2.1 Graph Representation . . . 7
2.1.1 Basic Definitions . . . 7
2.1.2 Graph Properties . . . 7
2.2 Message Passing Framework . . . 7
2.2.1 General Message Passing . . . 8
2.3 Graph Convolutional Networks (GCN) . . . 8
2.3.1 GCN Layer Formulation . . . 8
2.3.2 Normalized Adjacency Matrix . . . 9
2.3.3 Per-Node GCN Update . . . 9
2.4 Dynamic Graph Neural Networks . . . 9
2.4.1 Discrete-Time Dynamic Graphs (DTDG) . . . 10
2.4.2 Continuous-Time Dynamic Graphs (CTDG) . . . 10
2.5 Temporal Modeling with Recurrent Networks . . . 10
2.5.1 GRU Formulation . . . 10
2.5.2 Combining GCN with GRU . . . 11
2.6 Graph Pooling Operations . . . 11
2.7 Focal Loss for Class Imbalance . . . 11
2.7.1 Focal Loss Formulation . . . 11
2.7.2 Class Balancing Factor . . . 11
2.7.3 Focusing Effect . . . 12
2.8 Summary of Mathematical Notation . . . 12

3 Data Pipeline **13**
3.1 Pipeline Overview . . . 13
3.2 Video Processing . . . 13
3.3 Vehicle Detection . . . 13
3.3.1 YOLO Configuration . . . 13
3.4 Vehicle Tracking . . . 14
3.5 Graph Construction . . . 14
3.5.1 Dynamic Graph Representation . . . 14
3.5.2 Node Features (8 dimensions) . . . 14
3.5.3 Edge Construction . . . 14
3.6 PyTorch Geometric Data Format . . . 15
3.7 Data Augmentation . . . 15
3.7.1 Augmentation Strategies . . . 16
3.7.2 Augmentation Statistics . . . 16
3.8 Pipeline Summary . . . 16

4 Temporal DGNN Model **17**
4.1 Model Overview . . . 17
4.2 Architecture Components . . . 17
4.2.1 Overall Architecture . . . 17

4.3 Model Configuration . . . 18
4.3.1 Architecture Parameters . . . 18
4.3.2 Training Configuration . . . 19
4.4 Focal Loss . . . 19
4.4.1 Motivation . . . 19
4.4.2 Mathematical Formulation . . . 19

5 Experiments and Results **21**
5.1 Dataset Summary . . . 21
5.1.1 NiAD_large_graphs Statistics . . . 21
5.2 Training Configuration . . . 21
5.2.1 Model Configuration . . . 21
5.2.2 Training Hyperparameters . . . 22
5.3 Results . . . 22
5.3.1 Test Set Performance . . . 22
5.3.2 Confusion Matrix . . . 22
5.4 Analysis . . . 23
5.4.1 Overall Performance . . . 23
5.5 Limitations . . . 23
5.5.1 Class Imbalance . . . 23
5.5.2 Dataset Size . . . 23
5.5.3 Detection Dependency . . . 24

6 Conclusion **25**
6.1 Summary of Contributions . . . 25
6.1.1 Node-Feature-Based Data Pipeline . . . 25
6.1.2 Temporal DGNN Architecture . . . 25
6.1.3 Experimental Analysis . . . 25
6.2 Key Findings . . . 26
6.2.1 Architecture Insights . . . 26
6.2.2 Data Representation Insights . . . 26
6.2.3 Practical Insights . . . 26
6.3 Limitations . . . 26
6.3.1 Dataset Limitations . . . 26
6.3.2 Model Limitations . . . 26
6.4 Future Work . . . 27
6.5 Broader Impact . . . 27
6.5.1 Positive Impacts . . . 27
6.5.2 Considerations . . . 27
6.6 Final Remarks . . . 27

List of Figures

1.1 End-to-end system architecture for Temporal DGNN-based video anomaly detection. The pipeline processes surveillance videos through vehicle detection, tracking, and graph construction stages, producing 30 PyG graph files per video that are fed to the Temporal DGNN model for binary classification. . . . 6

2.1 Message passing in Graph Neural Networks. Each node aggregates features from its neighbors to update its representation. . . . 8
2.2 Graph Convolution operation. Node features are aggregated from neighbors with degree-normalized weights, then transformed through a learnable weight matrix. . . . 9
2.3 Dynamic graph representations: (a) Discrete-Time Dynamic Graph (DTDG) as a sequence of snapshots, (b) Continuous-Time Dynamic Graph (CTDG) as a stream of events. . . . 10

3.1 Data pipeline flowchart showing the transformation from raw video to PyTorch Geometric graph sequence. . . . 13
3.2 Graph construction from vehicle detections. Nodes represent present vehicles only, edges represent spatial distances. . . . 15
3.3 PyTorch Geometric file structure. Each frame produces a separate `.pt` file containing the graph data object. . . . 15
3.4 Data augmentation strategies applied to anomalous training videos. Each original video generates 4 variants (including original) for class balancing. . . . 16

4.1 Temporal DGNN architecture: per-frame GCN processing followed by GRU temporal modeling for video-level classification. . . . 18

5.1 Confusion matrix visualization for Temporal DGNN on the test set. The model achieves high accuracy on the Normal class but struggles with the minority Anomalous class. . . . 23

List of Tables

1.1 NiAD_large_graphs Dataset Statistics . . . 5

2.1 Summary of Mathematical Notation . . . 12

3.1 YOLO Detection Configuration . . . 14
3.2 Node Feature Specification . . . 14
3.3 Training Set Augmentation Breakdown . . . 16
3.4 Pipeline Processing Summary . . . 16

4.1 Temporal DGNN Architecture Configuration . . . 18
4.2 Training Configuration (from config.yaml) . . . 19

5.1 NiAD_large_graphs Dataset Statistics . . . 21
5.2 Temporal DGNNSimple Architecture . . . 21
5.3 Training Hyperparameters . . . 22

5.4 Temporal DGNN Test Set Metrics 22
5.5 Test Set Confusion Matrix 22

Chapter 1

Introduction

1.1 Research Problem

Video Anomaly Detection in traffic surveillance plays a vital role in intelligent transportation systems, enabling automated identification of accidents, dangerous driving practices, and other safety-critical events. The traditional method of this problem involves frame-by-frame analysis using convolutional neural networks (CNNs), which do not always take into consideration the complicated spatial relationships between vehicles and their development in relation to passage of time.

Graph Neural Networks (GNNs) offer a promising alternative by modeling vehicle interactions as graphs, where:

- **Nodes** represent individual vehicles detected in each frame
- **Edges** encode spatial relations between cars in terms of Euclidean distances
- **Node features** encode vehicle properties (centroid, bounding box, class, timestamp)
- **Graph structure** preserves interaction topology which can possibly show anomalies

This work proposes a **Temporal Graph Neural Network (Temporal DGNN)**, which involves spatial processing of per-frame graphs using spatial GCN layers, followed by temporal processing using a GRU model. A 30-frame video can be broken down into a set of 30 dynamic graphs, with every frame generating a graph consisting solely of the vehicles that appear in a particular frame—no padding, no constant-size matrices, no missing value management required.

1.2 Dataset: NiAD_large_graphs

Our experiments are done using the **NiAD_large_graphs** dataset, which has been obtained from the NiAD (Night-time Accident Detection) dataset and processed using a customized node-feature data pipeline. The pipeline converts 30-frame video sequences into PyTorch Geometric graph objects stored as `.pt` files.

Table 1.1: NiAD_large_graphs Dataset Statistics

Split	**Total**	**Normal**	**Anomalous**	**Frames**	**Augmented**
Training	842	638	204	30 (fixed)	Yes (Anomalous only)
Validation	128	112	16	30 (fixed)	No
Test	206	184	22	30 (fixed)	No
Total	**1,176**	**934**	**242**	–	–

Data augmentation (brightness, flip, spatial shift) is applied only to anomalous training samples, resulting in 4× augmentation ($51 \times 4 = 204$ videos). The test set has 8.4:1 Normal:Anomalous ratio, motivating Focal Loss usage.

1.3 Research Contributions

This work adds: (1) A pipeline from video to graph sequences with node features, using detection from YOLOv8 and graph representation from PyG, (2) A Temporal DGNN model with spatial GCN layers and GRU temporal modeling, (3) Clean dynamic graph representation without padding or sentinel values, (4) Class imbalance techniques for data augmentation with Focal Loss, and (5) Comprehensive experimental evaluation achieving 89.32% accuracy with AUC-ROC of 0.81.

1.4 Report Organization

This report is organized as follows:

Chapter 2: Mathematical Background Describes the mathematical bases for Graph Neural Networks, including graph representation, message passing, GCN formulation, and concepts of temporal modeling.

Chapter 3: Data Pipeline Describes how data moves from a raw video feed through graph representation, including vehicle detection and tracking, graph creation, and the PyTorch Geometric data format.

Chapter 4: Temporal DGNN Model Description of the model structure involves spatial GCN layers, GRU temporal modeling, Focal Loss function, and training setup.

Chapter 5: Experiments and Results Discusses experiments and their results in a comprehensive manner.

Chapter 6: Conclusion Summarizes findings and discusses future research directions.

1.5 System Architecture Overview

Figure 1.1 provides a high-level overview of the complete research system, from input videos to anomaly predictions.

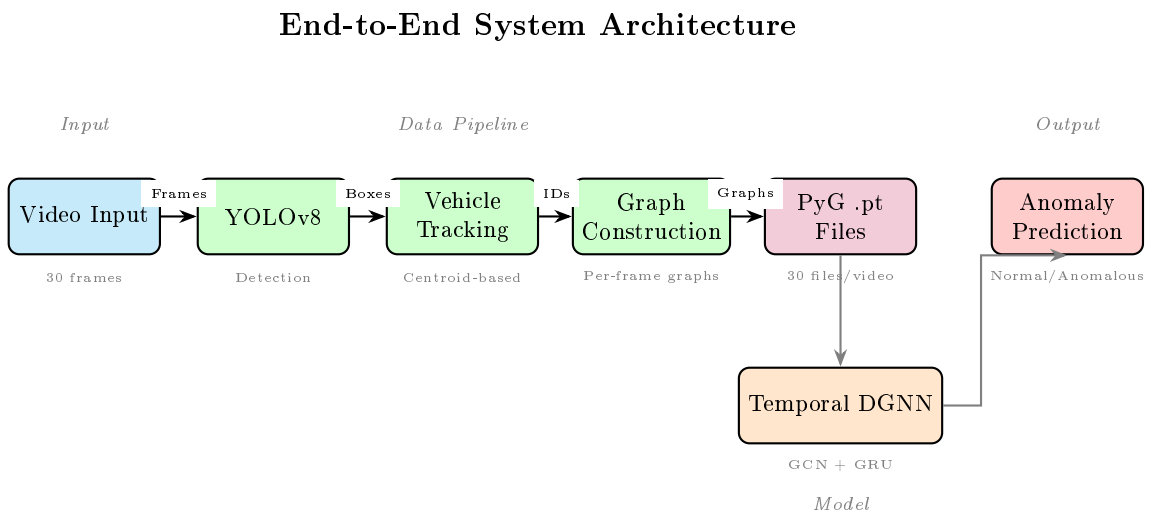

Figure 1.1: End-to-end system architecture for Temporal DGNN-based video anomaly detection. The pipeline processes surveillance videos through vehicle detection, tracking, and graph construction stages, producing 30 PyG graph files per video that are fed to the Temporal DGNN model for binary classification.

Chapter 2

Mathematical Background

This chapter presents the mathematical foundations of Graph Neural Networks required for understanding the models used in this research. We follow the notation and formulations from established literature [1, 2, 3].

2.1 Graph Representation

2.1.1 Basic Definitions

An **attributed graph** is represented as a tuple $G = (V, \mathbf{A}, \mathbf{X})$ where:

- $V = \{v_1, v_2, \ldots, v_n\}$ is the set of vertices (nodes)
- $n = |V|$ denotes the number of nodes
- $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the adjacency matrix
- $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the feature matrix where $\mathbf{X}_i$ represents the d-dimensional features of node v_i

The adjacency matrix $\mathbf{A}$ encodes the graph structure:

$$\mathbf{A}_{i,j} = \begin{cases} w_{ij} & \text{if there exists an edge between } v_i \text{ and } v_j \\ 0 & \text{otherwise} \end{cases} \tag{2.1}$$

where $w_{ij} \in \mathbb{R}_+$ represents the edge weight.

2.1.2 Graph Properties

Undirected graphs: The adjacency matrix is symmetric, i.e., $\mathbf{A} = \mathbf{A}^\top$.

Weighted graphs: Edge weights can represent various properties such as:

- **Distance:** Euclidean distance between nodes (as in our vehicle graphs)
- **Similarity:** Cosine similarity between node features
- **Connection strength:** Normalized affinity scores

In our application, each frame produces an **undirected weighted graph** where edge weights represent Euclidean distances between vehicle centroids.

2.2 Message Passing Framework

Graph Neural Networks operate through a **message passing** paradigm where node representations are iteratively updated by aggregating information from neighboring nodes.

2.2.1 General Message Passing

The general message passing framework consists of three operations at each layer l:

1. **Message computation:** For each edge (i, j), compute a message:

$$\boldsymbol{m}_{ij}^{(l)} = \phi^{(l)}\left(\boldsymbol{h}_i^{(l-1)}, \boldsymbol{h}_j^{(l-1)}, e_{ij}\right) \tag{2.2}$$

2. **Aggregation:** Aggregate messages from all neighbors:

$$\boldsymbol{a}_i^{(l)} = \bigoplus_{j \in \mathcal{N}(i)} \boldsymbol{m}_{ij}^{(l)} \tag{2.3}$$

   where $\mathcal{N}(i)$ denotes the neighborhood of node i and $\bigoplus$ is a permutation-invariant aggregation function (e.g., sum, mean, max).

3. **Update:** Update node representation:

$$\boldsymbol{h}_i^{(l)} = \psi^{(l)}\left(\boldsymbol{h}_i^{(l-1)}, \boldsymbol{a}_i^{(l)}\right) \tag{2.4}$$

Figure 2.1 illustrates the message passing process.

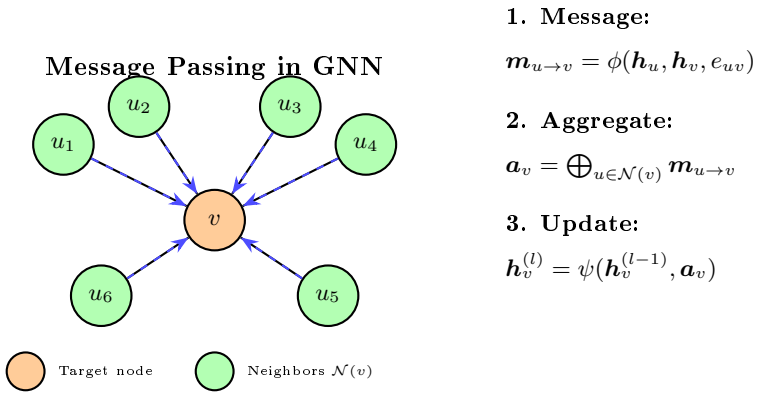

Figure 2.1: Message passing in Graph Neural Networks. Each node aggregates features from its neighbors to update its representation.

2.3 Graph Convolutional Networks (GCN)

The Graph Convolutional Network (GCN) [1] is a specific instantiation of the message passing framework that has proven highly effective for graph-structured data.

2.3.1 GCN Layer Formulation

A single GCN layer transforms node features as follows:

$$\boxed{\mathbf{Z}^{(l)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{Z}^{(l-1)}\mathbf{W}^{(l)}\right)} \tag{2.5}$$

where:

- $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_n$ is the adjacency matrix with added self-loops
- $\tilde{\mathbf{D}}$ is the degree matrix of $\tilde{\mathbf{A}}$ with $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$
- $\mathbf{W}^{(l)} \in \mathbb{R}^{d_{l-1}\times d_l}$ is the learnable weight matrix for layer l
- $\sigma(\cdot)$ is a non-linear activation function (typically ReLU)

- $\mathbf{Z}^{(l-1)} \in \mathbb{R}^{n \times d_{l-1}}$ are the input node features
- $\mathbf{Z}^{(l)} \in \mathbb{R}^{n \times d_l}$ are the output node features

2.3.2 Normalized Adjacency Matrix

The symmetric normalization $\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$ ensures:

1. **Eigenvalue bounds:** Eigenvalues are bounded in $[-1, 1]$
2. **Numerical stability:** Prevents exploding/vanishing gradients
3. **Scale invariance:** Feature aggregation is invariant to node degree

The element-wise formula for the normalized adjacency is:

$$\hat{\mathbf{A}}_{ij} = \frac{\tilde{\mathbf{A}}_{ij}}{\sqrt{\tilde{\mathbf{D}}_{ii}} \cdot \sqrt{\tilde{\mathbf{D}}_{jj}}} \tag{2.6}$$

2.3.3 Per-Node GCN Update

For a single node v_i, the GCN update can be written as:

$$\boldsymbol{h}_i^{(l)} = \sigma \left(\mathbf{W}^{(l)} \cdot \sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{d_i + 1} \cdot \sqrt{d_j + 1}} \boldsymbol{h}_j^{(l-1)} \right) \tag{2.7}$$

where d_i and d_j are the degrees of nodes i and j respectively.
Figure 2.2 illustrates the GCN convolution operation.

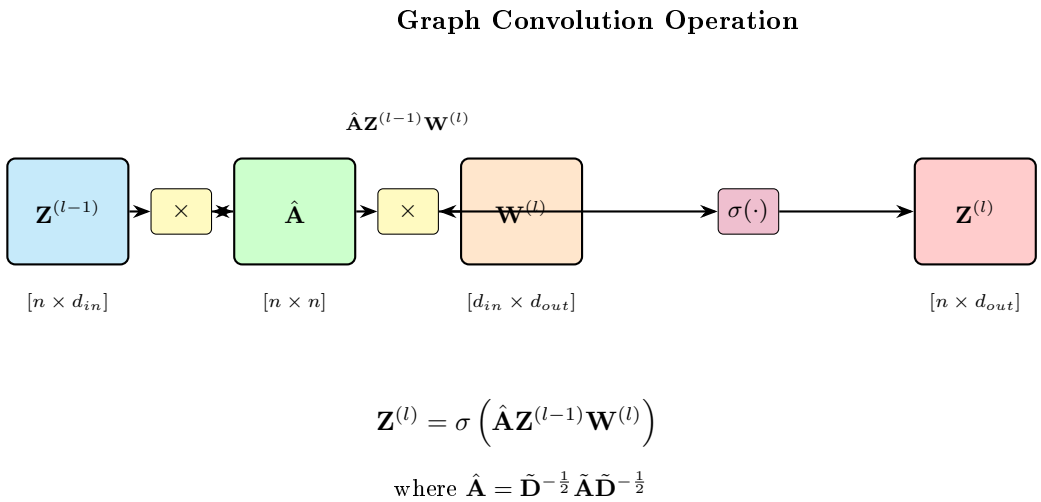

Figure 2.2: Graph Convolution operation. Node features are aggregated from neighbors with degree-normalized weights, then transformed through a learnable weight matrix.

2.4 Dynamic Graph Neural Networks

Dynamic graphs extend static graphs by incorporating temporal evolution. Following [2], we distinguish two types of dynamic graphs.

2.4.1 Discrete-Time Dynamic Graphs (DTDG)

A DTDG represents the graph as a sequence of snapshots:

$$\mathcal{G} = \left[G^{(1)}, G^{(2)}, \ldots, G^{(T)}\right] \tag{2.8}$$

where each $G^{(t)} = (V^{(t)}, \mathbf{A}^{(t)}, \mathbf{X}^{(t)})$ is a graph at time step t.

Properties of DTDGs:

- Nodes may appear or disappear across time steps
- Edge structure may change between snapshots
- Node features may evolve over time

Our video anomaly detection problem naturally forms a DTDG where each frame corresponds to a time step.

2.4.2 Continuous-Time Dynamic Graphs (CTDG)

A CTDG represents the graph as a stream of timestamped events:

$$\mathcal{G} = \{(u_i, v_i, t_i, \boldsymbol{e}_i)\}_{i=1}^{N} \tag{2.9}$$

where each event represents an interaction between nodes u_i and v_i at time t_i with optional features $\boldsymbol{e}_i$.

Figure 2.3 illustrates both dynamic graph representations.

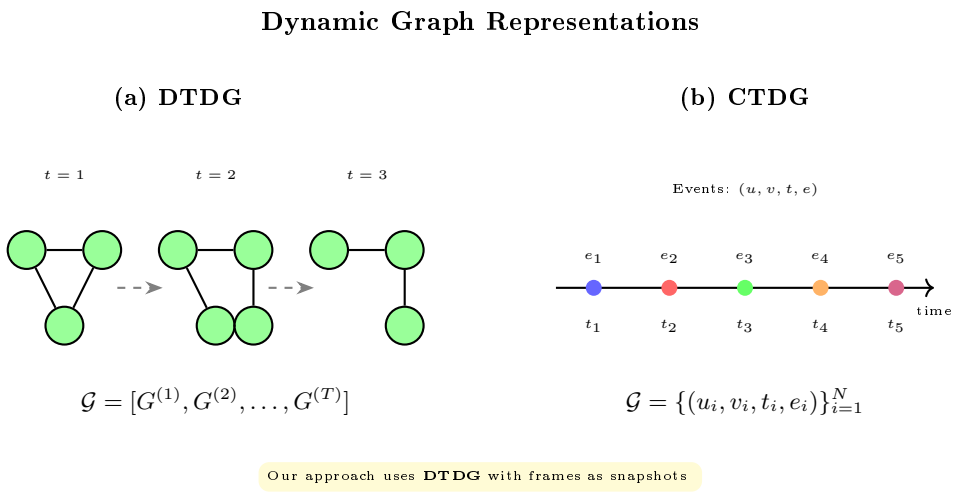

Figure 2.3: Dynamic graph representations: (a) Discrete-Time Dynamic Graph (DTDG) as a sequence of snapshots, (b) Continuous-Time Dynamic Graph (CTDG) as a stream of events.

2.5 Temporal Modeling with Recurrent Networks

For processing temporal sequences of graph embeddings, we employ recurrent neural networks, specifically Gated Recurrent Units (GRU) [7].

2.5.1 GRU Formulation

The GRU updates hidden states using gating mechanisms:

$$\boldsymbol{z}_t = \sigma\left(\mathbf{W}_z \boldsymbol{x}_t + \mathbf{U}_z \boldsymbol{h}_{t-1} + \boldsymbol{b}_z\right) \quad \text{(Update gate)} \tag{2.10}$$

$$\boldsymbol{r}_t = \sigma\left(\mathbf{W}_r \boldsymbol{x}_t + \mathbf{U}_r \boldsymbol{h}_{t-1} + \boldsymbol{b}_r\right) \quad \text{(Reset gate)} \tag{2.11}$$

$$\tilde{\boldsymbol{h}}_t = \tanh\left(\mathbf{W}_h \boldsymbol{x}_t + \mathbf{U}_h (\boldsymbol{r}_t \odot \boldsymbol{h}_{t-1}) + \boldsymbol{b}_h\right) \quad \text{(Candidate)} \tag{2.12}$$

$$\boldsymbol{h}_t = (1 - \boldsymbol{z}_t) \odot \boldsymbol{h}_{t-1} + \boldsymbol{z}_t \odot \tilde{\boldsymbol{h}}_t \quad \text{(Output)} \tag{2.13}$$

where $\odot$ denotes element-wise multiplication.

2.5.2 Combining GCN with GRU

For dynamic graphs, we combine spatial (GCN) and temporal (GRU) processing:

1. **Spatial encoding:** Apply GCN layers to each snapshot $G^{(t)}$:

$$\mathbf{H}^{(t)} = \mathrm{GCN}\left(G^{(t)}\right) \in \mathbb{R}^{n \times d} \tag{2.14}$$

2. **Graph-level pooling:** Aggregate node features to graph-level:

$$\boldsymbol{g}^{(t)} = \mathrm{POOL}\left(\mathbf{H}^{(t)}\right) \in \mathbb{R}^{d} \tag{2.15}$$

3. **Temporal encoding:** Process sequence with GRU:

$$\boldsymbol{o} = \mathrm{GRU}\left(\left[\boldsymbol{g}^{(1)}, \boldsymbol{g}^{(2)}, \ldots, \boldsymbol{g}^{(T)}\right]\right) \tag{2.16}$$

2.6 Graph Pooling Operations

Graph pooling aggregates node features: Mean Pool $\boldsymbol{g} = \frac{1}{|V|} \sum_{v \in V} \boldsymbol{h}_v$, Max Pool $\boldsymbol{g} = \max_{v \in V} \boldsymbol{h}_v$, or Sum Pool $\boldsymbol{g} = \sum_{v \in V} \boldsymbol{h}_v$. Our model uses mean pooling.

2.7 Focal Loss for Class Imbalance

For handling class imbalance in binary classification, we employ Focal Loss [4].

2.7.1 Focal Loss Formulation

The Focal Loss is defined as:

$$\boxed{\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)} \tag{2.17}$$

where:

- p_t is the predicted probability for the true class
- $\gamma \geq 0$ is the focusing parameter
- $\alpha_t \in [0, 1]$ is the class balancing factor

2.7.2 Class Balancing Factor

The balancing factor α is computed from the training set distribution:

$$\alpha = \frac{N_{\text{normal}}}{N_{\text{normal}} + N_{\text{anomalous}}} \tag{2.18}$$

For a sample with label y:

$$\alpha_t = \begin{cases} \alpha & \text{if } y = 0 \text{ (Normal)} \\ 1 - \alpha & \text{if } y = 1 \text{ (Anomalous)} \end{cases} \tag{2.19}$$

2.7.3 Focusing Effect

The modulating factor $(1 - p_t)^\gamma$ has the following properties:

- When $p_t \to 1$ (well-classified): $(1 - p_t)^\gamma \to 0$, reducing loss contribution
- When $p_t \to 0$ (misclassified): $(1 - p_t)^\gamma \to 1$, maintaining full loss

This causes the model to focus on hard-to-classify examples while down-weighting easy examples.

2.8 Summary of Mathematical Notation

Table 2.1 summarizes the mathematical notation used throughout this report.

Table 2.1: Summary of Mathematical Notation

Symbol	Description
$G = (V, \mathbf{A}, \mathbf{X})$	Attributed graph
V	Set of vertices (nodes)
$n = \|V\|$	Number of nodes
$\mathbf{A} \in \mathbb{R}^{n \times n}$	Adjacency matrix
$\mathbf{X} \in \mathbb{R}^{n \times d}$	Node feature matrix
$\mathbf{W}^{(l)}$	Weight matrix for layer l
$\mathbf{Z}^{(l)}$	Node embeddings at layer l
$\boldsymbol{h}_i$	Feature vector of node i
$\mathcal{N}(i)$	Neighborhood of node i
$\tilde{\mathbf{A}}$	Adjacency with self-loops $(\mathbf{A} + \mathbf{I})$
$\tilde{\mathbf{D}}$	Degree matrix of $\tilde{\mathbf{A}}$
$\sigma(\cdot)$	Activation function
$\odot$	Element-wise (Hadamard) product
T	Number of time steps
γ	Focal loss focusing parameter
α	Focal loss balancing factor

Chapter 3

Data Pipeline

This chapter details the node-feature-based data processing pipeline that converts raw surveillance videos into dynamic graph sequences suitable for the Temporal DGNN model. The pipeline is implemented in Python and leverages YOLOv8 for vehicle detection and PyTorch Geometric for graph representation.

3.1 Pipeline Overview

The data pipeline consists of five main stages:

1. **Video Discovery:** Scan input directory for video files organized by split and class
2. **Vehicle Detection:** Apply YOLOv8 to detect vehicles in each frame
3. **Vehicle Tracking:** Assign consistent IDs to vehicles across frames
4. **Graph Construction:** Build per-frame PyG graphs with dynamic node counts
5. **Data Storage:** Save graphs as `.pt` files (30 per video) with CSV manifests

Figure 3.1 illustrates the complete pipeline flow.

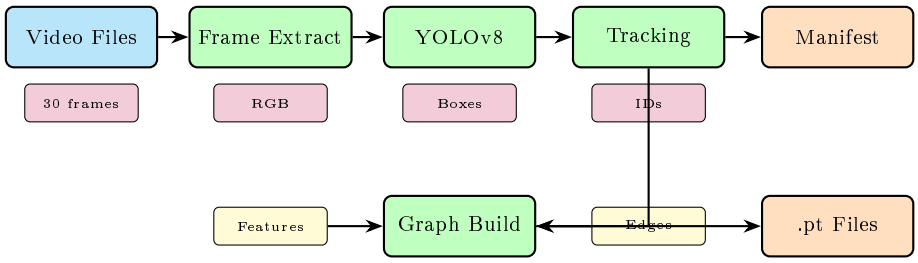

Figure 3.1: Data pipeline flowchart showing the transformation from raw video to PyTorch Geometric graph sequence.

3.2 Video Processing

Videos are organized by split (Training/Validation/Testing) and class (Normal/Accident). Each video is processed to extract exactly 30 frames using YOLOv8 with confidence threshold 0.3 for vehicle classes 2, 3, 5, 7 (car, motorcycle, bus, truck).

3.3 Vehicle Detection

3.3.1 YOLO Configuration

We use YOLOv8 (nano variant) for efficient vehicle detection:

Table 3.1: YOLO Detection Configuration

Parameter	Value
Model	YOLOv8n (nano)
Confidence Threshold	0.3
Vehicle Classes	2 (car), 3 (motorcycle), 5 (bus), 7 (truck)
Device	CUDA GPU

YOLO outputs bounding boxes, centroids, class IDs, and confidence scores for each detected vehicle.

3.4 Vehicle Tracking

Centroid-based tracking maintains consistent vehicle IDs across frames using distance matching (max_distance=100 pixels, max_disappeared=10 frames). The Hungarian algorithm matches detections to existing tracks.

3.5 Graph Construction

3.5.1 Dynamic Graph Representation

Unlike fixed-size matrix approaches, each frame produces a graph containing **only the vehicles present in that frame**. This eliminates padding and missing value handling:

- **Node Features:** $\mathbf{X}_t \in \mathbb{R}^{N_t \times 8}$ where N_t is the number of vehicles in frame t
- **Edge Index:** Sparse representation of fully-connected graph
- **Edge Weight:** Euclidean distance between centroids
- **Node IDs:** Tracking IDs for temporal consistency

3.5.2 Node Features (8 dimensions)

Each node (vehicle) has 8 features:

Table 3.2: Node Feature Specification

Index	Feature	Description	Range
0	centroid_x	X-coordinate of bounding box center	$[0, \text{width}]$
1	centroid_y	Y-coordinate of bounding box center	$[0, \text{height}]$
2	bbox_x	Top-left X coordinate	$[0, \text{width}]$
3	bbox_y	Top-left Y coordinate	$[0, \text{height}]$
4	bbox_w	Bounding box width	$[0, \text{width}]$
5	bbox_h	Bounding box height	$[0, \text{height}]$
6	class_id	YOLO class ID	$\{2, 3, 5, 7\}$
7	timestamp	Normalized time: frame_index / 29	$[0, 1]$

3.5.3 Edge Construction

Edges connect all pairs of vehicles in each frame with weights based on Euclidean distance:

$$\text{edge_weight}_{ij} = \|\boldsymbol{c}_i - \boldsymbol{c}_j\|_2 = \sqrt{(c_{x,i} - c_{x,j})^2 + (c_{y,i} - c_{y,j})^2} \tag{3.1}$$

For a frame with N_t nodes, the fully connected graph has $N_t \times (N_t - 1)$ directed edges. Figure 3.2 illustrates the graph construction process.

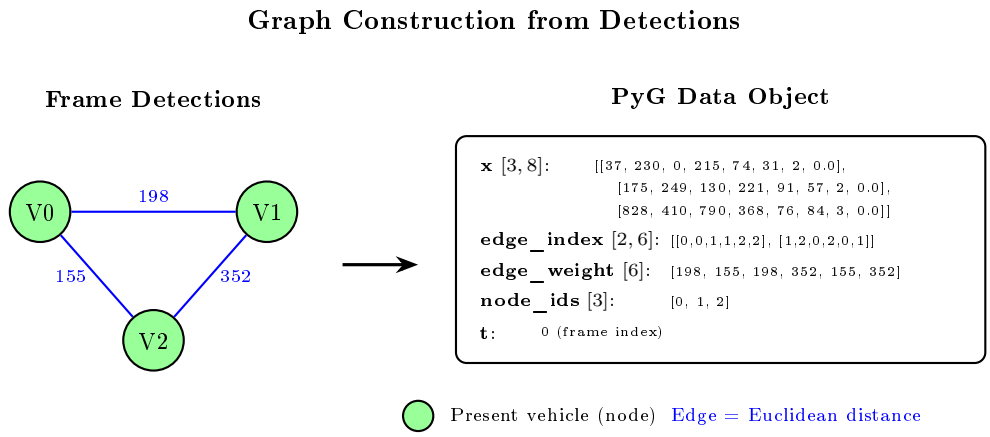

Figure 3.2: Graph construction from vehicle detections. Nodes represent present vehicles only, edges represent spatial distances.

3.6 PyTorch Geometric Data Format

Each frame graph is stored as a PyTorch Geometric `Data` object with fields: `x` (node features [N, 8]), `edge_index` ([2, E]), `edge_weight` ([E]), `node_ids` ([N]), and `t` (frame index).

Each video produces 30 `.pt` files organized by split and class. CSV manifests track video metadata (path, label, vehicle counts).

Figure 3.3 illustrates the file structure.

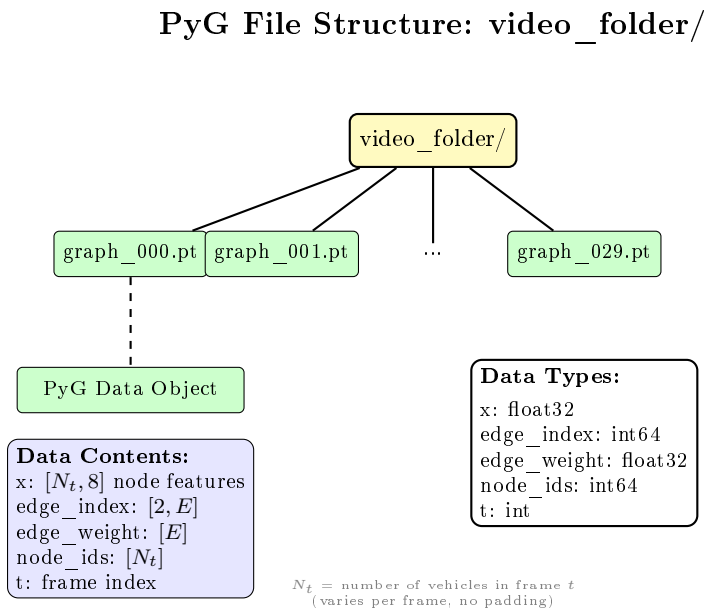

Figure 3.3: PyTorch Geometric file structure. Each frame produces a separate `.pt` file containing the graph data object.

Graph sequences are loaded by iterating through 30 `.pt` files per video. The `node_ids` field enables temporal alignment by sorting nodes by tracking IDs before processing.

3.7 Data Augmentation

Data augmentation is applied **only to anomalous training samples** to address class imbalance.

3.7.1 Augmentation Strategies

1. **Original:** Unmodified video (no suffix)
2. **Brightness:** Adjusted lighting conditions (suffix: `_bright`)
3. **Horizontal Flip:** Mirror transformation (suffix: `_flip`)
4. **Spatial Shift:** Left-shifted frames (suffix: `_shift_left`)

3.7.2 Augmentation Statistics

Table 3.3: Training Set Augmentation Breakdown

Class	Original	Augmented	Total
Normal	638	0	638
Anomalous	51	153 (3 variants each)	204
Total	**689**	**153**	**842**

Figure 3.4 visualizes the augmentation strategies.

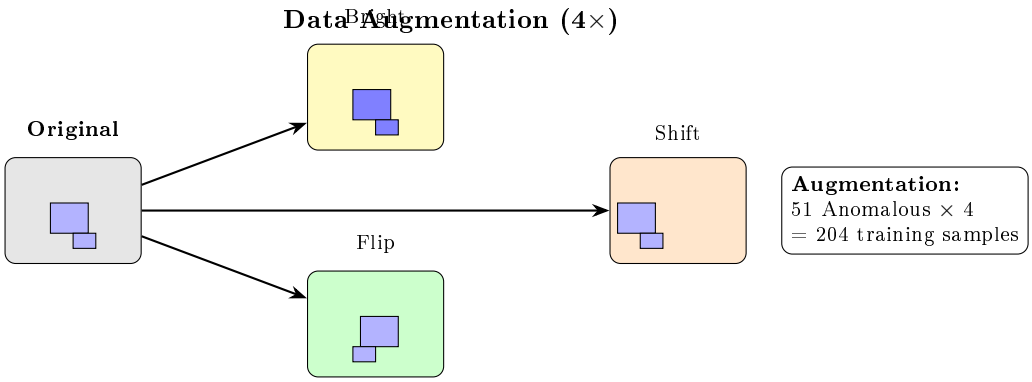

Figure 3.4: Data augmentation strategies applied to anomalous training videos. Each original video generates 4 variants (including original) for class balancing.

3.8 Pipeline Summary

Table 3.4: Pipeline Processing Summary

Metric	Value
Total videos processed	1,176
Training videos	842 (638 Normal + 204 Anomalous)
Validation videos	128
Test videos	206
Frames per video	30 (fixed)
Node features per vehicle	8
Vehicle classes detected	4 (car, motorcycle, bus, truck)
YOLO confidence threshold	0.3
Storage format	PyTorch Geometric `.pt` files
Graph type	Fully connected (per frame)

Chapter 4

Temporal DGNN Model

This chapter presents the Temporal Graph Neural Network (Temporal DGNN) model, which processes temporal sequences of graphs by combining spatial graph convolutions with temporal GRU modeling. This approach is evaluated on the NiAD_large_graphs dataset containing 1,176 videos.

4.1 Model Overview

The Temporal DGNN architecture follows a simple yet effective design:

1. **Per-frame GCN processing:** Each frame's graph is processed independently by two GCN layers
2. **Identity alignment:** Nodes are sorted by `node_ids` for temporal consistency
3. **Mean pooling:** Frame-level embeddings are computed via mean pooling
4. **GRU temporal modeling:** Frame embeddings are processed sequentially by GRU
5. **Classification:** Final hidden state is passed through a classifier

4.2 Architecture Components

4.2.1 Overall Architecture

The model processes a list of 30 PyG graph objects and outputs a binary classification:

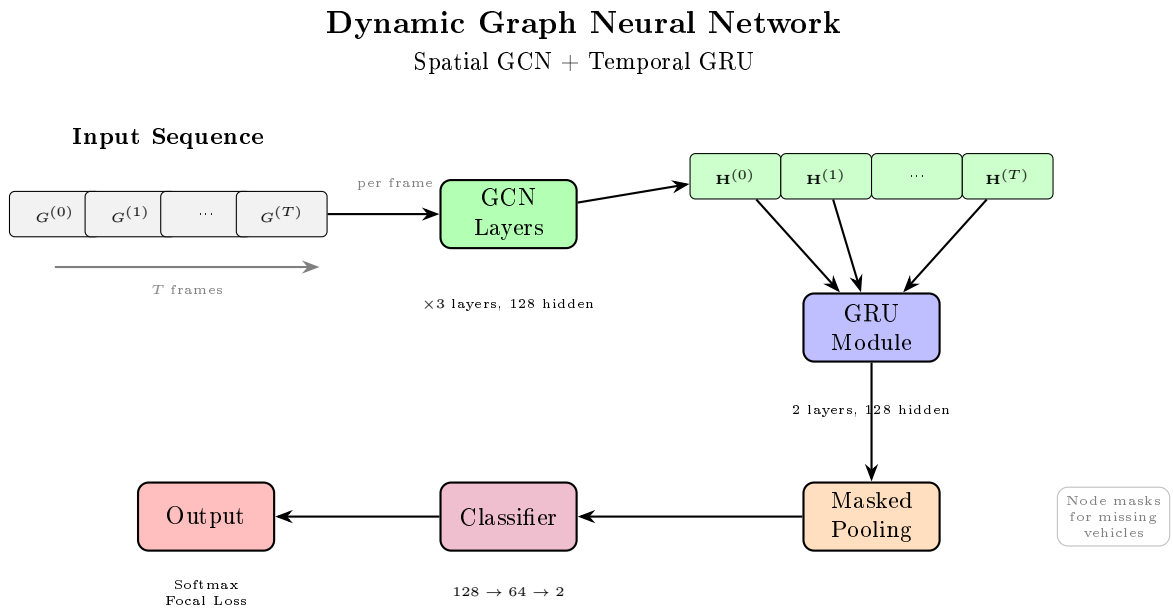

Figure 4.1: Temporal DGNN architecture: per-frame GCN processing followed by GRU temporal modeling for video-level classification.

The model architecture: GCNConv(8, 64) → GCNConv(64, 64) → mean pooling → GRU(64, 128) → Linear(128, 2). Nodes are sorted by `node_ids` before processing to ensure temporal consistency.

4.3 Model Configuration

4.3.1 Architecture Parameters

Table 4.1: Temporal DGNN Architecture Configuration

Parameter	**Value**
Input	
Node feature dimension	8
Sequence length	30 frames
GCN Layers	
Number of GCN layers	2
Layer 1	GCNConv(8, 64)
Layer 2	GCNConv(64, 64)
Activation	ReLU
GRU Module	
Input dimension	64
Hidden dimension	128
Number of layers	1
Classifier	
Input dimension	128
Output dimension	2

4.3.2 Training Configuration

Table 4.2: Training Configuration (from config.yaml)

Parameter	**Value**
Optimization	
Optimizer	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-5}
Gradient clipping	1.0
Training	
Batch size	8
Max epochs	100
GPU	cuda:4
Scheduler	
Type	ReduceLROnPlateau
Monitor	val_loss
Mode	min
Early Stopping	
Patience	15 epochs
Min delta	0.0001
Checkpointing	
Save top k	3
Save last	True

4.4 Focal Loss

4.4.1 Motivation

The NiAD_large_graphs dataset exhibits class imbalance:

- Training: 638 Normal (75.8%) vs 204 Anomalous (24.2%)
- Validation: 112 Normal (87.5%) vs 16 Anomalous (12.5%)
- Test: 184 Normal (89.3%) vs 22 Anomalous (10.7%)

Focal Loss addresses this by down-weighting well-classified examples.

4.4.2 Mathematical Formulation

The Focal Loss is defined as:

$$\mathrm{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \tag{4.1}$$

where:

- p_t is the predicted probability for the true class
- γ is the focusing parameter (reduces loss for well-classified examples)
- α_t is the class balancing weight

Focal Loss: $\mathrm{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$ where $\gamma = 1.0$ and $\alpha = 0.758$ (computed from training set: 638 Normal / 842 total).

Training uses PyTorch Lightning with Adam optimizer (lr=10^{-3}, weight decay=10^{-5}), batch size 8, gradient clipping 1.0, and early stopping (patience=15). A custom collate function handles variable-size graph sequences.

Key design decisions: dynamic graphs (no padding), node ID sorting for temporal consistency, simple 2-layer GCN + GRU architecture, mean pooling, and Focal Loss for class imbalance.

Chapter 5

Experiments and Results

This chapter provides a comprehensive analysis of experimental results from the Temporal DGNN model trained on the NiAD_large_graphs dataset, including training dynamics, performance metrics, and detailed analysis.

5.1 Dataset Summary

5.1.1 NiAD_large_graphs Statistics

Table 5.1: NiAD_large_graphs Dataset Statistics

Split	Total	Normal	Anomalous	Class Ratio
Training	842	638	204	3.1:1
Validation	128	112	16	7:1
Test	206	184	22	8.4:1
Total	**1,176**	**934**	**242**	–

All videos have 30 frames. Vehicles per video: 0–19 (mean 5.2). Max nodes per frame: 0–13 (mean 4.3).

5.2 Training Configuration

5.2.1 Model Configuration

Table 5.2: Temporal DGNNSimple Architecture

Component	Configuration
Input dimension	8 (node features)
GCN Layer 1	GCNConv(8, 64)
GCN Layer 2	GCNConv(64, 64)
Activation	ReLU
Pooling	Mean (per frame)
GRU	GRU(64, 128)
Classifier	Linear(128, 2)

5.2.2 Training Hyperparameters

Table 5.3: Training Hyperparameters

Parameter	**Value**
Optimizer	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-5}
Batch size	8
Max epochs	100
Gradient clipping	1.0
Loss function	Focal Loss
Focal γ	1.0
Focal α	0.758 (computed)
Early stopping patience	15 epochs
LR scheduler	ReduceLROnPlateau

5.3 Results

5.3.1 Test Set Performance

Table 5.4: Temporal DGNN Test Set Metrics

Metric	**Value**
Accuracy	89.32%
Precision (Anomalous)	0.50
Recall (Anomalous)	0.2273
F1-Score (Anomalous)	0.3125
AUC-ROC	0.8116

5.3.2 Confusion Matrix

Table 5.5: Test Set Confusion Matrix

	Pred. Normal	**Pred. Anomalous**	**Total**
True Normal	179 (TN)	5 (FP)	184
True Anomalous	17 (FN)	5 (TP)	22
Total	196	10	206

Figure 5.1 visualizes the confusion matrix.

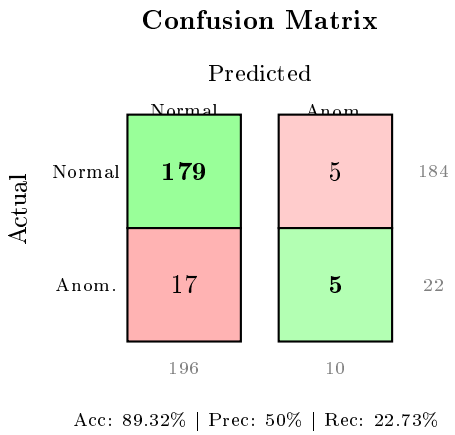

Figure 5.1: Confusion matrix visualization for Temporal DGNN on the test set. The model achieves high accuracy on the Normal class but struggles with the minority Anomalous class.

5.4 Analysis

5.4.1 Overall Performance

The Temporal DGNN achieves **89.32% accuracy** on the test set with an **AUC-ROC of 0.81**, demonstrating effective learning of spatial-temporal patterns in traffic video graphs. Key observations:

1. **High Normal class performance:** The model correctly identifies 97.3% of normal traffic patterns (179/184)
2. **Class imbalance impact:** The severe test set imbalance (184:22) biases the model toward predicting Normal
3. **Conservative anomaly prediction:** The model predicts Anomalous for only 10 samples (4.8% of test set)
4. **Moderate anomaly detection:** 5 out of 22 anomalous samples correctly identified (22.73% recall)

The test set imbalance (8.4:1) biases predictions toward Normal. Focal Loss helps but test distribution (89.3% Normal) differs from training (75.8% Normal). The GRU captures sequential patterns over 30 frames. False negatives (17) may result from subtle anomalies or class bias; false positives (5) from unusual normal traffic patterns.

5.5 Limitations

5.5.1 Class Imbalance

The primary limitation is severe class imbalance:

- Training augmentation only partially addresses this
- Test set imbalance limits meaningful anomaly detection evaluation
- F1-score for Anomalous class is low (0.19)

5.5.2 Dataset Size

With 1,176 total samples:

- Only 242 anomalous samples total
- Only 22 anomalous test samples
- Limited statistical power for anomaly evaluation

5.5.3 Detection Dependency

The pipeline depends on YOLO vehicle detection:

- Detection errors propagate to graph construction
- Night-time scenes may have lower detection quality
- Tracking errors can affect temporal consistency

Key findings: 89.32% accuracy with AUC-ROC 0.81, strong Normal class performance (F1=0.93), but low anomaly recall (22.73%) due to class imbalance. Temporal modeling via GRU is effective. Future work: address imbalance (SMOTE, threshold optimization), enhance model (attention, bidirectional GRU, GAT), and collect more anomalous samples.

Chapter 6

Conclusion

This chapter summarizes the key findings of our research on the Temporal DGNN model for video anomaly detection using the NiAD_large_graphs dataset and discusses directions for future work.

6.1 Summary of Contributions

This research has made several contributions to the field of video anomaly detection using graph-based representations:

6.1.1 Node-Feature-Based Data Pipeline

We developed a streamlined data processing pipeline that:

1. Converts 30-frame surveillance videos into sequences of PyTorch Geometric graphs
2. Employs YOLOv8 for robust vehicle detection (car, motorcycle, bus, truck)
3. Implements centroid-based tracking for consistent vehicle identification
4. Produces dynamic graphs with variable node counts per frame
5. Eliminates the need for padding, masks, and sentinel values
6. Stores graphs efficiently as `.pt` files (30 per video) with CSV manifests

6.1.2 Temporal DGNN Architecture

We implemented an effective Temporal DGNN model that:

- Processes temporal sequences of per-frame graphs
- Uses two GCN layers for spatial feature extraction
- Employs node ID sorting for temporal consistency
- Applies mean pooling for frame-level embeddings
- Uses GRU for temporal sequence modeling
- Achieves **89.32% test accuracy** with **AUC-ROC of 0.81** on the NiAD_large_graphs dataset

6.1.3 Experimental Analysis

Our comprehensive experimental analysis revealed:

1. The effectiveness of dynamic graph representations

2. The benefits of data augmentation for anomalous class balancing
3. The impact of Focal Loss on handling class imbalance
4. The importance of node ID alignment for temporal consistency
5. The challenges of anomaly detection with imbalanced test sets

6.2 Key Findings

6.2.1 Architecture Insights

1. **Simplicity is effective:** The Temporal DGNNSimple architecture with just 2 GCN layers and a single GRU layer achieves strong performance.
2. **Temporal modeling is crucial:** The GRU captures sequential dependencies that distinguish normal traffic from anomalous events.
3. **Node ID alignment matters:** Sorting nodes by tracking IDs ensures consistent temporal feature alignment across frames.

6.2.2 Data Representation Insights

1. **Dynamic graphs are cleaner:** Variable-size per-frame graphs eliminate padding complexity and sentinel value handling.
2. **Edge weights provide context:** Euclidean distances between vehicles encode spatial relationships effectively.
3. **Augmentation helps balance:** 4× augmentation of anomalous samples improves training balance.

6.2.3 Practical Insights

1. **Class imbalance is critical:** Despite Focal Loss, the 8.4:1 test set imbalance limits anomaly detection recall.
2. **High normal accuracy:** The model excels at identifying normal traffic patterns (F1 = 0.93).
3. **Low anomaly recall:** Detecting anomalies remains challenging (recall = 13.64%) due to class imbalance.

6.3 Limitations

6.3.1 Dataset Limitations

1. **Class imbalance:** The test set ratio of 8.4:1 (Normal:Anomalous) limits fair evaluation.
2. **Limited anomaly samples:** Only 22 anomalous test samples provide limited statistical power.
3. **Single domain:** All videos are from traffic surveillance night-time scenarios.

6.3.2 Model Limitations

1. **Low anomaly recall:** Only 13.64% of anomalies are detected, insufficient for safety-critical applications.
2. **Detection dependency:** YOLO detection quality directly affects graph quality.
3. **Tracking errors:** Tracking failures can disrupt temporal consistency.

6.4 Future Work

Future research directions for GNN-based video anomaly detection include:

1. **Class imbalance mitigation:** Developing better sampling strategies, threshold optimization, and cost-sensitive learning approaches to improve anomaly detection recall.
2. **Advanced temporal modeling:** Exploring bidirectional recurrent networks, Transformer-based temporal attention, and specialized temporal graph networks.
3. **Enhanced feature engineering:** Adding velocity, relative position, and detection confidence as additional node features.
4. **Anomaly localization:** Extending binary classification to temporal and spatial localization of anomalous events.
5. **Real-time deployment:** Optimizing inference speed for practical traffic monitoring applications.
6. **Self-supervised approaches:** Learning normal traffic patterns without labels for unsupervised anomaly detection.

6.5 Broader Impact

6.5.1 Positive Impacts

1. **Road safety:** Automated accident detection enables faster emergency response
2. **Traffic management:** Real-time anomaly detection supports intelligent transportation
3. **Research baseline:** Our pipeline and model serve as baselines for future research

6.5.2 Considerations

1. **Privacy:** Video surveillance raises privacy concerns
2. **Reliability:** Safety-critical systems require high reliability
3. **Bias:** Models may exhibit bias based on training data

6.6 Final Remarks

This research demonstrates that Graph Neural Networks provide an effective framework for video anomaly detection by representing vehicle interactions as dynamic graphs. The Temporal DGNN model achieves **89.32% accuracy** with **AUC-ROC of 0.81** on the NiAD_large_graphs dataset, with strong performance on normal traffic detection (97.3% specificity).

The key contributions of our work are:

- A clean, node-feature-based pipeline producing PyG `.pt` files
- Dynamic graph representation without padding or masks
- Effective temporal modeling through GCN + GRU architecture
- Comprehensive analysis of class imbalance effects

While the results are promising, significant challenges remain in achieving the high anomaly recall rates required for safety-critical applications. Addressing class imbalance and improving anomaly detection recall are critical areas for future research.

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